

1. (10 分) 计算二重积分:

$$\iint_D \ln(1+x^2+y^2) dx dy, \quad D: x^2+y^2 \leq 1, x \geq 0, y \geq 0.$$

2. (10 分) 计算三重积分:

$$\iiint_{\Omega} (y^2+z^2) dV, \quad \Omega: 0 \leq z \leq x^2+y^2 \leq 1.$$

3. (10 分) 设曲线 C 为椭圆 $x^2/16+y^2/9=1$ 沿逆时针方向。计算曲线积分:

$$\oint_C \frac{x dy - y dx}{x^2 + y^2}.$$

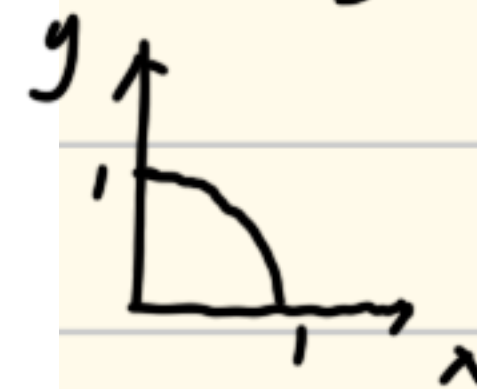
4. (10 分) 计算曲面积分:

$$\iint_S (x^2 y^2 + y^2 z^2 + z^2 x^2) dS,$$

其中 S 为圆锥面 $z = \sqrt{x^2+y^2}$ 被柱面 $x^2+y^2=1$ 截下部分。

$$\therefore x = r \cos \theta, y = r \sin \theta$$

$$\iint_D r \cdot \ln(1+r^2) \cdot r dr d\theta$$



$$\int_0^{\frac{\pi}{2}} d\theta \int_0^1 r \ln(1+r^2) dr$$

$$= \frac{\pi}{2} \cdot \int_0^1 r \ln(1+r^2) dr$$

$$\int_0^1 r \ln(1+r^2) dr = \frac{1}{2} \int_0^1 \ln(1+r^2) dr^2 = \frac{1}{2} \cdot \int_0^1 \ln(1+r) dr$$

$$= \frac{1}{2} (r \ln(1+r)) \Big|_0^1 - \int_0^1 r \cdot d \ln(1+r)$$

$$= \frac{1}{2} (\ln 2 - \int_0^1 \frac{r}{1+r} dr)$$

$$1 - \frac{1}{1+r}$$

$$\ln 2 - [r - \ln(1+r)] \Big|_0^1$$

$$= \ln 2 - (1 - \ln 2) = 2 \ln 2 - 1$$

$$= \frac{\pi}{2} (2 \ln 2 - 1)$$

$$\therefore \text{原式} = \frac{\pi}{4} (2 \ln 2 - 1)$$

2.

注意看区域!

关于 xOz, yOz 面对称, 关于 x, y 为偶函数, 仍考虑第一象限

$$x = r \cos \theta, y = r \sin \theta, z = z$$

$$4 \int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^1 dr \cdot \int_0^{r^2} r \cdot (r^2 \sin^2 \theta + z^2) dz$$

$$= 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^3 \sin^2 \theta \cdot (1-r^2) + \frac{1}{3} (1-r^2)^3 r dr$$

$$\text{令 } r = \sin \alpha,$$

$$\text{则 } \int_0^1 r^3 \sin^2 \theta (1-r^2) + \frac{1}{3} (1-r^2)^3 r dr$$

$$= \int_0^{\frac{\pi}{2}} (\sin^3 \alpha \cos \alpha \sin^2 \theta + \frac{1}{3} \cos^3 \alpha) \cos \alpha \sin \alpha d\alpha$$

$$= \int_0^{\frac{\pi}{2}} \sin^3 \alpha \cos^2 \alpha \sin^2 \theta + \frac{1}{3} \cos^4 \alpha d\alpha$$

$$= \frac{2}{15} \sin^2 \theta + \frac{1}{3} \cdot \frac{2}{2} \cdot \frac{3}{8} = \frac{2}{15} \sin^2 \theta + \frac{2}{16}$$

$$\therefore \text{原式} = 4 \int_0^{\frac{\pi}{2}} (\frac{2}{15} \sin^2 \theta + \frac{2}{16}) d\theta$$

$$= 4 \cdot (\frac{2}{15} \cdot \frac{\pi}{4} + \frac{\pi^2}{32}) = \frac{2}{15} \pi + \frac{\pi^2}{8}$$

$$4 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 dr \int_0^{r^2} (r^3 \sin^2 \theta + r \cdot z^2) dz$$

$$4 \cdot \int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^1 r^5 \sin^2 \theta + \frac{r}{3} r^6 dr$$

$$= 4 \cdot \int_0^{\frac{\pi}{2}} (\frac{1}{6} \sin^2 \theta + \frac{1}{24}) d\theta$$

$$= 4 \cdot (\frac{1}{6} \cdot \frac{\pi}{4} + \frac{\pi}{24}) = \frac{\pi}{4}$$

$$\frac{2}{3} - \frac{8}{15}$$

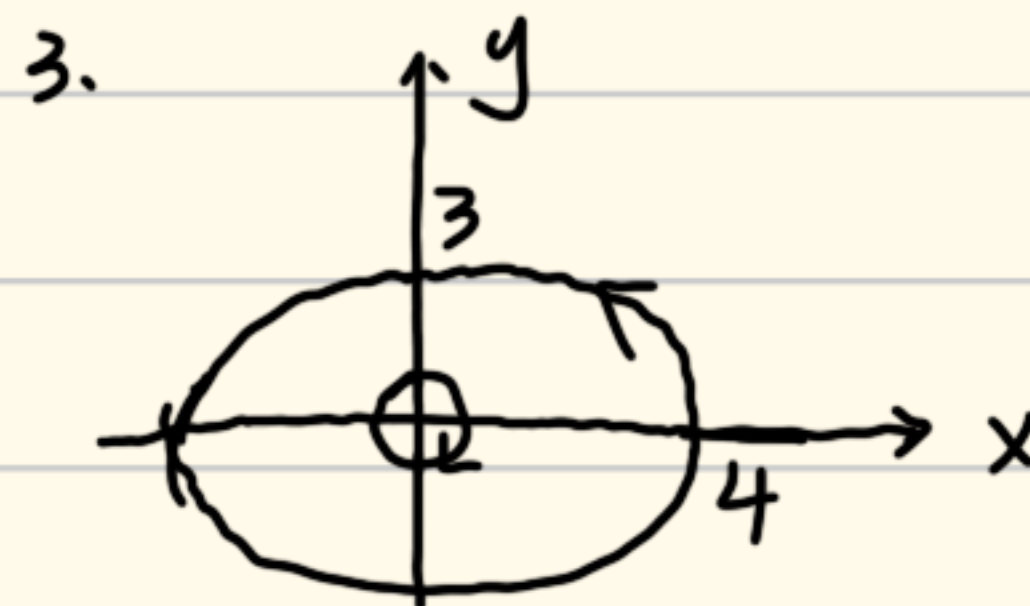
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$$\iint_S (x^2y^2 + y^2z^2 + z^2x^2) dS,$$

其中 S 为圆锥面 $z = \sqrt{x^2 + y^2}$ 被柱面 $x^2 + y^2 = 1$ 截下部分。



设区域 $D: x^2 + y^2 \leq \varepsilon^2$, C_1^+ 为 D 的边界顺时针方向, C 所围区域为 D ($\varepsilon > 0$)

由格林公式,

$$\oint_{C-C_1^+} \frac{xdy - ydx}{x^2 + y^2} = \iint_{D-b_1^+} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) d\sigma = 0$$

$$\begin{aligned} \text{则 } \oint_C \frac{xdy - ydx}{x^2 + y^2} &= \int_{C-C_1^+} \frac{xdy - ydx}{x^2 + y^2} + \int_{C_1^+} \frac{xdy - ydx}{x^2 + y^2} = \int_{C_1^-} \frac{xdy - ydx}{x^2 + y^2} = \int_{C_1^-} \frac{xdy - ydx}{\varepsilon^2} \\ &= \frac{-1}{\varepsilon^2} \iint_{b_1} (-1-1) d\sigma \\ &= \frac{1}{\varepsilon^2} \cdot 2 \cdot \pi \cdot \varepsilon^2 = 2\pi \end{aligned}$$

故原式 = 2π

4. S 关于 xOy , yOz 面对称, 而关于 x, y 为偶函数.

则仅考虑第一象限即可, 记第一象限在 xOy 上投影为 D .

$$\text{由轮换对称, } \iint_S y^2 z^2 dS = \iint_S x^2 z^2 dS,$$

$$x = r \cos \theta, y = r \sin \theta, z = r.$$

$$x_r = \cos \theta, y_r = \sin \theta, z_r = 1$$

$$x_\theta = -r \sin \theta, y_\theta = r \cos \theta, z_\theta = 0$$

$$E = z, F = 0, G = r^2$$

$$dS = \sqrt{2} r^2 d\sigma = \sqrt{2} r d\sigma$$

$$\iint_S x^2 y^2 dS = \iint_D (r^4 \sin^2 \theta \cos^2 \theta) \cdot \sqrt{2} r \cdot dr d\theta$$

$$= \sqrt{2} \iint_D r^5 \sin^2 \theta \cos^2 \theta dr d\theta$$

$$= \sqrt{2} \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^5 \sin^2 \theta \cos^2 \theta dr$$

$$= \frac{\sqrt{2}}{6} \cdot \frac{\pi}{2} \left(\frac{1}{2} - \frac{3}{8} \right) = \frac{\sqrt{2}}{12} \cdot \frac{1}{8} \cdot \pi$$

$$\begin{aligned} \iint_S y^2 z^2 dS &= \iint_D \sqrt{2} r \cdot (r^4 \sin^2 \theta) dr d\theta = \sqrt{2} \iint_D r^5 \sin^2 \theta \cdot dr d\theta = \sqrt{2} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \int_0^1 r^5 dr \\ &= \frac{\sqrt{2}}{6} \cdot \frac{\pi}{4} \end{aligned}$$

$$\begin{aligned} \therefore \text{原式} &= \frac{\sqrt{2}}{24} \pi \times 4 \times 2 + \frac{\sqrt{2}}{12} \cdot \frac{1}{8} \pi \times 4 \\ &= \frac{\sqrt{2}}{3} \pi + \frac{\sqrt{2}}{24} \pi = \frac{9}{24} \sqrt{2} \pi \end{aligned}$$

$\frac{3}{8} \sqrt{2} \pi$ 约分!

5. (15 分) 计算曲面积分:

$$\oiint_S xdydz + ydzdx + zdxdy,$$

其中 S 为抛物面 $z = x^2 + y^2$ 被平面 $z = 4$ 所截部分的外侧.

6. (10 分) 求下面常微分方程的所有解: $y' = xy + 3x + 2y + 6$.

7. (15 分) 求下面常微分方程的通解: $y'' - 4y' + 3y - 4e^x = 0$.

8. (10 分) 设平面有界闭区域 D 为

$$D = \left\{ (x, y) \in \mathbb{R}^2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}, \quad a, b > 0.$$

设曲线 L 为 D 的边界, 函数 $P(x, y), Q(x, y)$ 在 D 上有连续的一阶偏导数. 记 $F = (P, Q)$, n 为曲线 L 的单位外法向量. 证明:

$$\oint_{L^+} F \cdot n \, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy.$$

$$6. \quad y' + (-x-2)y = 3x+6$$

解对应的齐次方程 $y' = (x+2)y$

$$\frac{dy}{dx} = (x+2)y,$$

①若 $y=0$, 则原式不成立舍去

$$\textcircled{2} \text{若 } y \neq 0, \quad \frac{dy}{y} = (x+2)dx. \quad \ln y = \frac{1}{2}x^2 + 2x + C$$

$$y = Ce^{\frac{1}{2}x^2 + 2x}, \quad \text{令 } C = \varphi(x).$$

$$y = \varphi(x) e^{\frac{1}{2}x^2 + 2x}$$

$$y' = \varphi'(x) \cdot e^{\frac{1}{2}x^2 + 2x} + \varphi(x) \cdot (x+2) \cdot e^{\frac{1}{2}x^2 + 2x}$$

$$\therefore \varphi'(x) \cdot e^{\frac{1}{2}x^2 + 2x} = 3x+6$$

$$\varphi'(x) = \frac{3x+6}{e^{\frac{1}{2}x^2 + 2x}},$$

$$\int \frac{3x+6}{e^{\frac{1}{2}x^2 + 2x}} = 3 \int \frac{1}{e^{\frac{1}{2}x^2 + 2x}} d(\frac{1}{2}x^2 + 2x)$$

$$= -3e^{-\frac{1}{2}x^2 - 2x} + C_1$$

$$\therefore y = -3 + C_1 e^{\frac{1}{2}x^2 + 2x}$$

$$5. \text{ 补充平面 } S_1^+ : \begin{cases} x^2 + y^2 = 4 \\ z = 4 \end{cases}, \text{ 指向上侧}$$

由高斯公式

$$\therefore \oiint_{S+S_1^+} xdydz + ydzdx + zdxdy$$

$$= \iiint_V 3 \cdot dV = 3 \cdot \int_0^{2\pi} d\theta \int_0^2 dr \cdot \int_{r^2}^4 r \cdot dz$$

$$= 3 \cdot 2\pi \cdot \int_0^2 r \cdot (4-r^2) \cdot dr$$

$$= 6\pi \int_0^2 (4-r^2) dr^2$$

$$= 3\pi \cdot (-\ln|4-r^2|) \Big|_0^{\sqrt{2}} ?$$

$$= \cancel{3\pi(-\ln 2 + \ln 4)} = 3\ln 2 \pi$$

$$4r-r^3$$

$$2r^2 - \frac{1}{4}r^4$$

$$8-4$$

$$4r^2 - \frac{1}{5}r^4$$

$$16-8=8$$

$$\text{而 } \oiint_S xdydz + ydzdx + zdxdy = (3\ln 2) \cdot 2 - \iiint_{S_1^+} xdydz + ydzdx + zdxdy$$

$$= 3\pi \ln 2 - \iiint_S 4 \cdot dS$$

$$= 3\pi \ln 2 - 4 \cdot 2 \cdot 4$$

$$= 3\pi \ln 2 - 16\pi$$

5. (15 分) 计算曲面积分:

$$\oiint_S x \mathrm{d}y \mathrm{d}z + y \mathrm{d}z \mathrm{d}x + z \mathrm{d}x \mathrm{d}y,$$

其中 S 为抛物面 $z = x^2 + y^2$ 被平面 $z = 4$ 所截部分的外侧.

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设曲线 L 为 D 的边界, 函数 $P(x, y), Q(x, y)$ 在 D 上有连续的一阶偏导数. 记 $\mathbf{F} = (P, Q)$, $\mathbf{n}$ 为曲线 L 的单位外法向量. 证明:

$$\oint_{L^+} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}s = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) \mathrm{d}x \mathrm{d}y.$$

$$7. \quad y'' - 4y' + 3y = 4e^x$$

$$\lambda^2 - 4\lambda + 3 = 0. \quad \lambda_1 = 1, \lambda_2 = 3, \quad \text{齐次通解为 } C_1 e^x + C_2 e^{3x}$$

1 是单特征根

$$\text{设 } y = ax e^x$$

$$y'' = a(x+2)e^x, \quad y' = a(x+1)e^x.$$

$$\therefore a(x+2-4x-4)e^x + 3ax e^x$$

$$= a \cdot -2 \cdot e^x = 4e^x.$$

$$a = -2.$$

$$\text{特解为 } y = -2x e^x$$

$$\therefore \text{通解为 } C_1 e^x + C_2 e^{3x} - 2x e^x$$

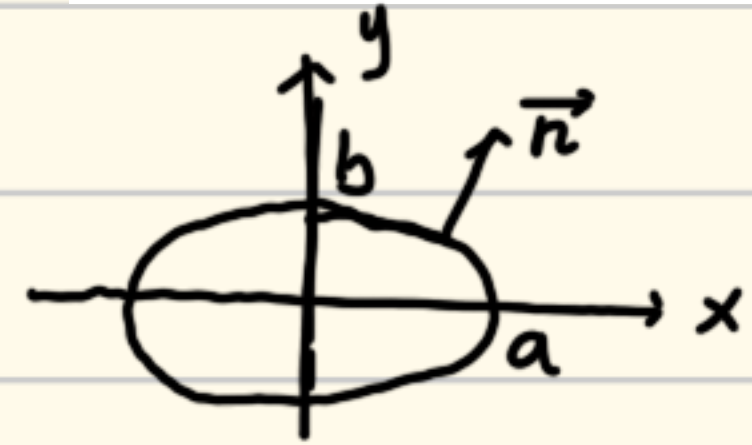
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$$\oint_{L^+} F \cdot n \, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy.$$



L 上点 (x, y) 上的法向量是 $(\frac{2x}{a^2}, \frac{2y}{b^2})$, 单位法向量是 $\frac{1}{\sqrt{\frac{x^2}{a^4} + \frac{y^2}{b^4}}} \cdot (\frac{x}{a^2}, \frac{y}{b^2})$.

则单位逆时针方向切向量是 $\frac{1}{\sqrt{\frac{x^2}{a^4} + \frac{y^2}{b^4}}} \cdot (\frac{y}{b^2}, -\frac{x}{a^2})$

$$\text{则 } \oint_{L^+} F \cdot n \, ds = \oint_{L^+} P \cdot dy - Q \cdot dx$$

$$\text{由格林公式, 原式} = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy$$

9. (10 分) 设 $f(x)$ 为 $\mathbb{R}$ 上的连续函数。证明:

$$\iint_S f(x+y+z) dS = 2\pi \int_{-1}^1 f(\sqrt{3}\xi) d\xi,$$

其中 S 为单位球面 $x^2 + y^2 + z^2 = 1$ 。

9. S 上的单位法向量是 $\frac{1}{\sqrt{x^2+y^2+z^2}} \cdot (x, y, z)$

考虑 $z > 0$ 部分, $z = \sqrt{1-x^2-y^2}$

$$2x + 2z \cdot z_x = 0, \quad z_x = -\frac{x}{z}, \quad \text{同理 } z_y = -\frac{y}{z}$$

$$\therefore ds = \sqrt{\frac{x^2+y^2}{z^2} + 1} dz = \frac{1}{z} dx dy$$

$$\text{原式} = 2 \iint_{x^2+y^2 \leq 1} f(x+y+\sqrt{1-x^2-y^2}) \cdot \frac{1}{z} dx dy$$